

Supplementary Material

PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk

GeroScience

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Numerical values are transcribed from the cited canonical artifacts and rounded only for display; source artifacts retain full precision.

Supplementary Table S1. Paired C-index contrasts in the temporal-validation cohort

Contrast	Cohort	ΔC (A – B)	95% CI low	95% CI high	Bootstrap proportion $\Delta C > 0$	Replicates	Seed
CalibratedTotalRisk – DemoRisk	temporal_test	0.01628	0.01109	0.02199	1.0	2000	20260821
CalibratedTotalRisk – PhenoAge NCP	temporal_test	0.00205	–0.00326	0.00748	0.776	2000	20260821
PhysioRisk – PhenoAge NCP acceleration	temporal_test	0.01625	–0.00049	0.03285	0.97	2000	20260821

Note: Paired percentile intervals were obtained from 2,000 participant bootstrap replicates in the temporal-validation cohort. Contrast estimates and interval limits are shown to the manuscript's five-decimal reporting precision; the canonical source artifact retains full precision.

Provenance: tables/cindex_bootstrap_contrasts.csv.

Supplementary Table S2. Proportional-hazards diagnostics

Model	Cohort	N	Deaths	Time transform	Test statistic	P value	Material violation	Flag rule	Cox warning	Scaled-residual region summary
DemoRisk	temporal_test	6,739	416	rank	2.925	0.08720586	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=-0.087; middle_37_72m: mean=0.044; late_73m_plus: mean=0.099
PhysioRisk	temporal_test	6,739	416	rank	2.467	0.11627106	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=-0.035; middle_37_72m: mean=0.013; late_73m_plus: mean=0.052
CaibratedTotalRisk	temporal_test	6,739	416	rank	16.539	0.00004765	True	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=-0.177; middle_37_72m: mean=0.108; late_73m_plus: mean=0.161
HematologicStress	temporal_test	6,739	416	rank	3.852	0.04967436	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	
RenalHepaticTurnover	temporal_test	6,739	416	rank	4.304	0.03802453	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	

Note: The prespecified descriptive flag was $p<0.01$ without multiplicity adjustment. The regional scaled-residual summaries are reported where present in the canonical artifact.

Provenance: tables/ph_diagnostics.csv.

Supplementary Table S3. Diagnostic log-time interaction analyses

Model	N	Deaths	Score scale	Time interaction	Reference time (months)	Log HR/SD at 60m	HR/SD at 12m	HR/SD at 60m	HR/SD at 96m	Interaction coefficient	HR multiplier/e-fold time	95% CI low	95% CI high	P value	Direction
DemoRisk	6,739	416	actual temporal-cohort SD	score_z * log(time_months/60)	60	1.358	3.399	3.890	4.047	0.084	1.087	−0.055	0.223	0.23794	effect_strengthens_over_time
PhysioRisk	6,739	416	actual temporal-cohort SD	score_z * log(time_months/60)	60	0.356	1.366	1.428	1.446	0.027	1.028	−0.032	0.087	0.36401	effect_strengthens_over_time
CalibratedTotalRisk	6,739	416	actual temporal-cohort SD	score_z * log(time_months/60)	60	1.320	2.842	3.744	4.057	0.171	1.187	0.066	0.277	0.00150	effect_strengthens_over_time

Note: These analyses are diagnostic only and did not replace or refit the fixed predictors used in temporal validation.

Provenance: tables/ph_time_interaction.csv.

Supplementary Table S4. Fixed-horizon calibration of CalibratedTotalRisk

Horizon (months)	Training baseline survival	Mean predicted mortality	KM observed mortality	Observed – predicted	O/E ratio	Global Cox slope	Global slope CI low	Global slope CI high	Horizon intercept	Horizon slope	IPCW Brier score	IPCW usable N	No temporal recalibration
60	0.983687	0.046101	0.046533	0.000432	1.009	0.811	0.762	0.859	–0.101	0.941	0.03776	4,709	True
96	0.966595	0.085710	0.088256	0.002546	1.030	0.811	0.762	0.859	0.067	0.999	0.06018	1,518	True

Note: Absolute risks used the development-derived coefficients and baseline survival. Horizon-specific logistic calibration and Brier scores used temporal-test censoring weights for evaluation only; no temporal recalibration occurred.

Provenance: tables/calibration_fixed_horizons.csv.

Supplementary Table S5. Censoring-robust discrimination diagnostics

Model	Evaluation cohort	Censoring-distribution cohort	Training N for censoring	Test N	Horizon (months)	Uno C (IPCW)	Cumulative/dynamic AUC (IPCW)	Implementation
DemoRisk	temporal_test	training (1999-2010)	13,090	6,739	60	0.823	0.834	scikit-survival 0.28.0
DemoRisk	temporal_test	training (1999-2010)	13,090	6,739	96	0.827	0.853	scikit-survival 0.28.0
PhysioRisk	temporal_test	training (1999-2010)	13,090	6,739	60	0.654	0.657	scikit-survival 0.28.0
PhysioRisk	temporal_test	training (1999-2010)	13,090	6,739	96	0.654	0.677	scikit-survival 0.28.0
CalibratedTotalRisk	temporal_test	training (1999-2010)	13,090	6,739	60	0.839	0.850	scikit-survival 0.28.0
CalibratedTotalRisk	temporal_test	training (1999-2010)	13,090	6,739	96	0.843	0.870	scikit-survival 0.28.0
PhenoAge NCP	temporal_test	training (1999-2010)	13,090	6,739	60	0.838	0.850	scikit-survival 0.28.0
PhenoAge NCP	temporal_test	training (1999-2010)	13,090	6,739	96	0.841	0.875	scikit-survival 0.28.0
PhenoAge NCP acceleration	temporal_test	training (1999-2010)	13,090	6,739	60	0.641	0.655	scikit-survival 0.28.0
PhenoAge NCP acceleration	temporal_test	training (1999-2010)	13,090	6,739	96	0.638	0.698	scikit-survival 0.28.0

Note: The development cohort estimated the censoring distribution. These were diagnostics of fixed temporal-test predictions, not model-training steps.

Provenance: tables/censoring_robust_discrimination.csv.

Supplementary Table S6. Cycle-specific preselection biomarker missingness and cohort provenance

Cycle start year	Biomarker	Mortality-eligible N	Missing N	Missing (%)	Denominator population
1999	LBXSAL	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXSCR	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXGLU	5,445	3,168	58.182	mortality_eligible_before_complete_case_restriction
1999	LBXLYPCT	5,445	762	13.994	mortality_eligible_before_complete_case_restriction
1999	LBXMCV	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
1999	LBXRDW	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
1999	LBXSAPSI	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXWBCSI	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
2001	LBXSAL	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXSCR	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXGLU	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXLYPCT	5,987	717	11.976	mortality_eligible_before_complete_case_restriction
2001	LBXMCV	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2001	LBXRDW	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2001	LBXSAPSI	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXWBCSI	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2003	LBXSAL	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXSCR	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXGLU	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXLYPCT	5,610	591	10.535	mortality_eligible_before_complete_case_restriction
2003	LBXMCV	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2003	LBXRDW	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2003	LBXSAPSI	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXWBCSI	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2005	LBXSAL	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXSCR	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXGLU	5,561	3,137	56.411	mortality_eligible_before_complete_case_restriction
2005	LBXLYPCT	5,561	557	10.016	mortality_eligible_before_complete_case_restriction
2005	LBXMCV	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2005	LBXRDW	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2005	LBXSAPSI	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXWBCSI	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2007	LBXSAL	6,219	655	10.532	mortality_eligible_before_complete_case_restriction
2007	LBXSCR	6,219	656	10.548	mortality_eligible_before_complete_case_restriction
2007	LBXGLU	6,219	3,468	55.765	mortality_eligible_before_complete_case_restriction
2007	LBXLYPCT	6,219	626	10.066	mortality_eligible_before_complete_case_restriction
2007	LBXMCV	6,219	612	9.841	mortality_eligible_before_complete_case_restriction
2007	LBXRDW	6,219	612	9.841	mortality_eligible_before_complete_case_restriction
2007	LBXSAPSI	6,219	657	10.564	mortality_eligible_before_complete_case_restriction
2007	LBXWBCSI	6,219	613	9.857	mortality_eligible_before_complete_case_restriction
2009	LBXSAL	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXSCR	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXGLU	6,510	3,581	55.008	mortality_eligible_before_complete_case_restriction
2009	LBXLYPCT	6,510	500	7.680	mortality_eligible_before_complete_case_restriction
2009	LBXMCV	6,510	491	7.542	mortality_eligible_before_complete_case_restriction
2009	LBXRDW	6,510	491	7.542	mortality_eligible_before_complete_case_restriction
2009	LBXSAPSI	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXWBCSI	6,510	491	7.542	mortality_eligible_before_complete_case_restriction

Cycle start year	Biomarker	Mortality-eligible N	Missing N	Missing (%)	Denominator population
2011	LBXSAL	5,849	700	11.968	mortality_eligible_before_complete_case_restriction
2011	LBXSCR	5,849	700	11.968	mortality_eligible_before_complete_case_restriction
2011	LBXGLU	5,849	3,254	55.633	mortality_eligible_before_complete_case_restriction
2011	LBXLYPCT	5,849	545	9.318	mortality_eligible_before_complete_case_restriction
2011	LBXMCV	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2011	LBXRDW	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2011	LBXSAPSI	5,849	702	12.002	mortality_eligible_before_complete_case_restriction
2011	LBXWBCSI	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2013	LBXSAL	6,100	482	7.902	mortality_eligible_before_complete_case_restriction
2013	LBXSCR	6,100	482	7.902	mortality_eligible_before_complete_case_restriction
2013	LBXGLU	6,100	3,377	55.361	mortality_eligible_before_complete_case_restriction
2013	LBXLYPCT	6,100	430	7.049	mortality_eligible_before_complete_case_restriction
2013	LBXMCV	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2013	LBXRDW	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2013	LBXSAPSI	6,100	483	7.918	mortality_eligible_before_complete_case_restriction
2013	LBXWBCSI	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2015	LBXSAL	5,974	600	10.044	mortality_eligible_before_complete_case_restriction
2015	LBXSCR	5,974	601	10.060	mortality_eligible_before_complete_case_restriction
2015	LBXGLU	5,974	3,406	57.014	mortality_eligible_before_complete_case_restriction
2015	LBXLYPCT	5,974	564	9.441	mortality_eligible_before_complete_case_restriction
2015	LBXMCV	5,974	563	9.424	mortality_eligible_before_complete_case_restriction
2015	LBXRDW	5,974	563	9.424	mortality_eligible_before_complete_case_restriction
2015	LBXSAPSI	5,974	601	10.060	mortality_eligible_before_complete_case_restriction
2015	LBXWBCSI	5,974	563	9.424	mortality_eligible_before_complete_case_restriction

Note: Missingness was quantified within each cycle among mortality-eligible participants before complete-case restriction. The denominator is repeated for each biomarker within a cycle; biomarker-specific missing counts are not joint complete-case exclusions.

Provenance: tables/missingness_by_cycle.csv.

Supplementary Table S7. Mortality-eligible participants included in versus excluded from the benchmark cohort

Group	Population definition	N	Age mean	Age SD	Female (%)	Deaths N	Deaths (%)	Median follow-up (months)
benchmark_included	Complete for notebook 01 EXTENDED_16	19,829	47.009	19.050	50.643	2,936	14.807	124
excluded_before_benchmark	Missing one or more notebook 01 EXTENDED_16 fields	33,426	47.784	19.862	52.486	6,168	18.453	122

Note: Inclusion was defined by completeness for notebook 01’s EXTENDED_16 fields. The two groups partition the 53,255 mortality-eligible participants: 19,829 included and 33,426 excluded.

Provenance: tables/cohort_included_excluded_characteristics.csv.

Supplementary Table S8. Common complete-case and training-derived-imputation sensitivity analyses

Analysis	Score	Cohort	N	Deaths	C-index	Age correlation	Missing-data rule	Status
common_complete_case	PhysioRisk	temporal_test	6,739	416	0.654	0.106	identical participants; all no-CRP comparator and PhysioRisk biomarkers observed	performed
common_complete_case	PhenoAge NCP (no-CRP proxy)	temporal_test	6,739	416	0.841	0.961	identical participants; all no-CRP comparator and PhysioRisk biomarkers observed	performed
common_training_derived_imputation	PhysioRisk	temporal_test	6,739	416	0.654	0.106	no-op in retained benchmark because all required biomarkers are observed; upstream imputation cannot be reconstructed	performed as degenerate no-missing sensitivity; extension to excluded source participants not estimable without frozen residualization parameters
common_training_derived_imputation	PhenoAge NCP (no-CRP proxy)	temporal_test	6,739	416	0.841	0.961	no-op in retained benchmark because all required biomarkers are observed; upstream imputation cannot be reconstructed	performed as degenerate no-missing sensitivity; extension to excluded source participants not estimable without frozen residualization parameters

Note: All 6,739 temporal-validation participants in the retained benchmark already had the biomarkers required for both scores. Common training-derived imputation was therefore a no-op and changed no observations; extension to excluded source participants was not estimable from the frozen retained scoring path.

Provenance: tables/comparator_sensitivity.csv.

Supplementary Table S9. Historical submitted benchmark results retained for provenance only

Status	Interpretation	Cohort	Risk score	C-index	HR per 1 SD	HR CI low	HR CI high	P value	Age correlation
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	BaselineRisk	0.827	3.736	3.347	4.170	4.38e-122	0.970
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	PhysioRisk_2axis	0.654	2.772	2.656	2.893	0	0.106
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	TotalRisk_2axis	0.840	1.877	1.815	1.942	1.08e-293	0.811
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	PhenoAge_noCRP_proxy	0.841	3.976	3.637	4.347	3.4e-202	0.961
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	PhenoAgeAccel	0.638	1.432	1.360	1.508	3.7e-42	0.033
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	BaselineRisk	0.855	4.286	4.097	4.484	0	0.971
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	PhysioRisk_2axis	0.602	1.334	1.295	1.374	1.07e-81	-0.000
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	TotalRisk_2axis	0.858	2.427	2.372	2.483	0	0.790
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	PhenoAge_noCRP_proxy	0.864	3.670	3.556	3.787	0	0.958
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	PhenoAgeAccel	0.586	1.246	1.219	1.274	1.13e-85	0.000

Note: Historical provenance only. The historical PhysioRisk hazard-ratio fit did not converge, and TotalRisk_2axis is the obsolete unit-weight composite. These rows are not current manuscript results.

Provenance: manuscript/generated/table_s9_historical_submitted_results.csv (generated by scripts/build_s4_manuscript_assets.py from tables/physiorisk_vs_phenoage_aligned_comparison.csv).